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scAnnotatR: Framework to accurately classify cell types in single-cell RNA-sequencing data

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Introduction

Single-cell RNA-sequencing has become a key tool for biomedical research. One of the main steps in analyzing single-cell RNA-sequencing data is to classify the observed cell types. While most existing tools show good sensitivity and specificity in classifying cell types, they often fail to adequately not-classify cells that are not present in the used reference and do not scale to the continuously increasing size of current scRNA-seq datasets. Therefore, we developed scAnnotatR to solve all mentioned issues. The package performs comparable accuracy, sensitivity and specificity to other available cell annotation methods while being able to reveal unknown populations. scAnnotatR also works on large datasets (up to more than 600,000 cells). In terms of functionality, scAnnotatR supports both Seurat and Bioconductor's SingleCellExperiment (SCE) objects, and provides user-friendly pipelines suitable for a wide range of users. scAnnotatR's prediction returns all detailed information that can be used in further analyses.

The package is available at: https://github.com/grisslab/scAnnotatR

Software design

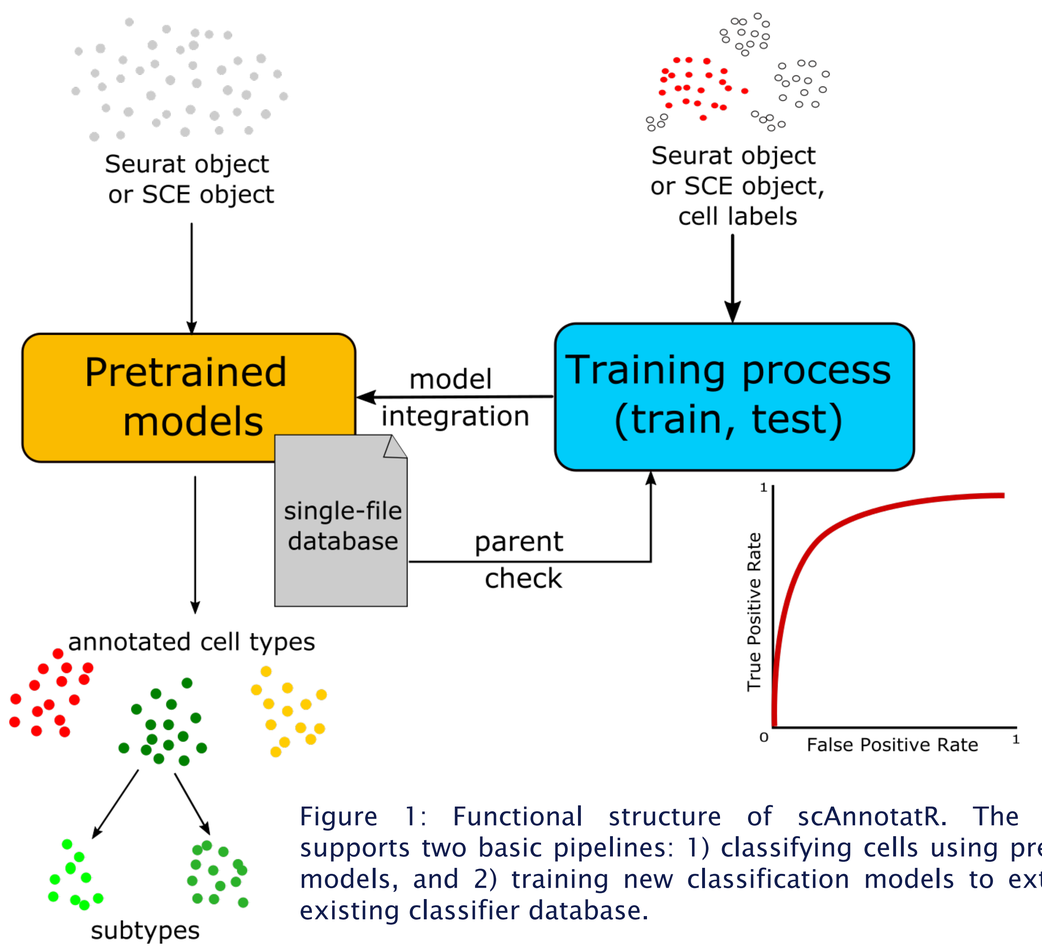

Figure 1: Functional structure of scAnnotatR. The package supports two basic pipelines: 1) classifying cells using pre-trained models, and 2) training new classification models to extend the existing classifier database.

Key features

- scAnnotatR classifies single cells at high accuracy.
- scAnnotatR is able to detect unknown population, because each classifier is for a specific cell type.
- scAnnotatR is the only cell classification tool having high accuracy that can scale to large datasets (up to more than 600,000 cells).
- The prediction probability threshold is changeable to increase/decrease model sensitivity and therefore decrease/increase model specificity.
- scAnnotatR produces detailed prediction output.

Results

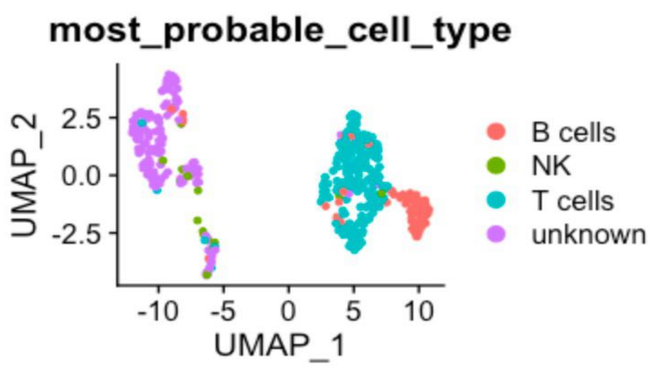

Figure 2: Example where scAnnotatR's prediction was projected onto an UMAP plot using the DimPlot function in Seurat package. scAnnotaR's prediction is directly integrated into the input (Seurat/SCE) object and can be reused in downstream analyses.

```
#>                                          seurat_clusters  B_cells_p      NK_p
#> Cy80_II_CD45_B10_S886_comb                             4 0.01164624 0.3178055
#> cy80.Cd45.pos.PD1.pos.B09.S45.comb                     2 0.01195767 0.3602068
#> cy80.Cd45.pos.Pd1.neg.S366.H06.S366.comb               3 0.99964045 0.4695694
#> cy80.Cd45.pos.Pd1.neg.S202.A10.S202.comb               3 0.98726305 0.2758547
#> cy80.Cd45.pos.Pd1.neg.S201.A09.S201.comb               3 0.99900954 0.2837854
#> cy80.Cd45.pos.Pd1.neg.S221.B05.S221.comb               3 0.99975201 0.4407576
#>                                          T_cells_p predicted_cell_type
#> Cy80_II_CD45_B10_S886_comb              0.07151094             unknown
#> cy80.Cd45.pos.PD1.pos.B09.S45.comb      0.88766329             T cells
#> cy80.Cd45.pos.Pd1.neg.S366.H06.S366.comb 0.04706935            B cells
#> cy80.Cd45.pos.Pd1.neg.S202.A10.S202.comb 0.12078281            B cells
#> cy80.Cd45.pos.Pd1.neg.S201.A09.S201.comb 0.13813295            B cells
#> cy80.Cd45.pos.Pd1.neg.S221.B05.S221.comb 0.03885957            B cells
#>                                          most_probable_cell_type     clust_pred
#> Cy80_II_CD45_B10_S886_comb                               unknown 92.86% unknown
#> cy80.Cd45.pos.PD1.pos.B09.S45.comb                       T cells 87.65% T cells
#> cy80.Cd45.pos.Pd1.neg.S366.H06.S366.comb                 B cells   100% B cells
#> cy80.Cd45.pos.Pd1.neg.S202.A10.S202.comb                 B cells   100% B cells
#> cy80.Cd45.pos.Pd1.neg.S201.A09.S201.comb                 B cells   100% B cells
#> cy80.Cd45.pos.Pd1.neg.S221.B05.S221.comb                 B cells   100% B cells
```

Figure 3: An illustration of detailed prediction output by scAnnotatR. scAnnotatR's prediction output consisting of each cell type prediction probability, the ambiguous prediction, the most probable predicted cell type, and the most dominant cell type in clusters (if cluster information was provided).

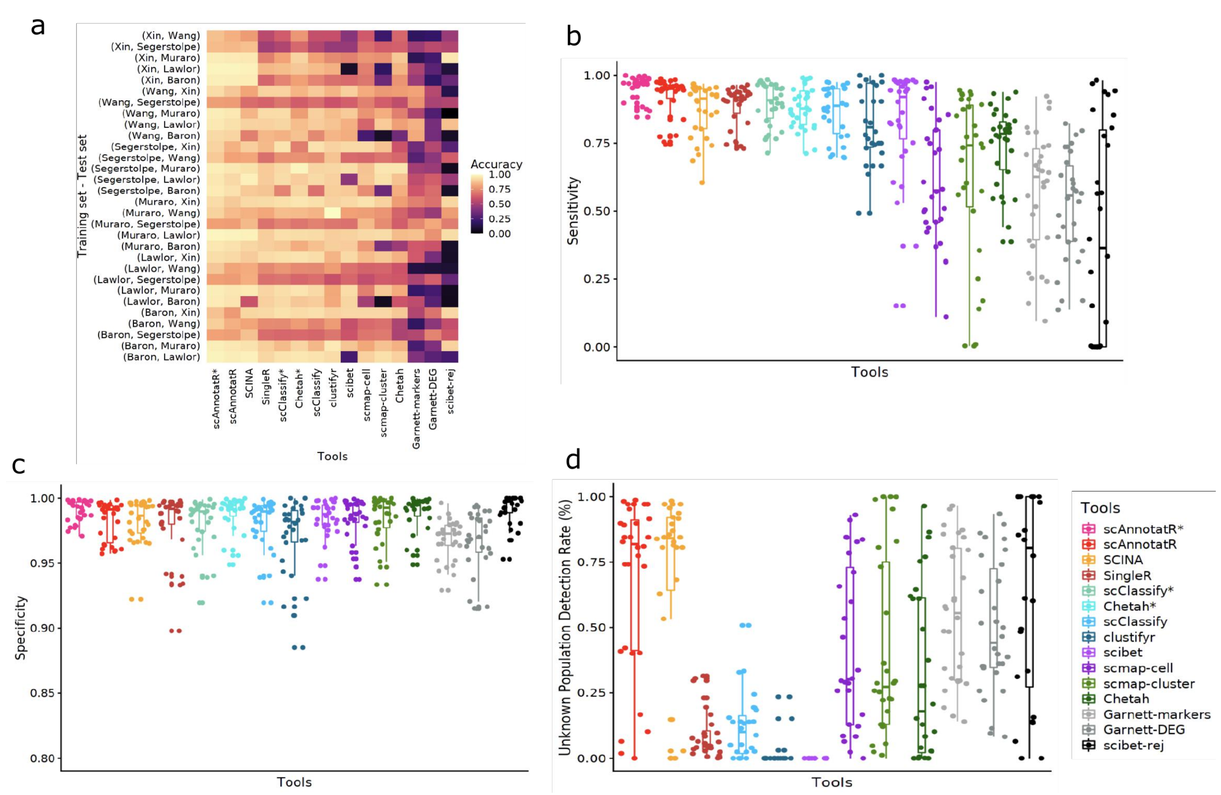

Figure 4: Benchmark results when classifying distinct cell populations in six pancreas datasets. The benchmark was performed in a six-fold cross-validation scheme where one dataset was used for training and the other five for testing. All metrics were ordered by the tools' median accuracy. scAnnotatR was consistently among the tools with the highest accuracy (a), sensitivity (b), and specificity (c) throughout all datasets while being able to correctly not-classify unknown cell types (d). SingleR, scClassify, SciBet (default settings), and clustifyr, showed high accuracy and sensitivity, but they failed in not-classifying unknown cells.

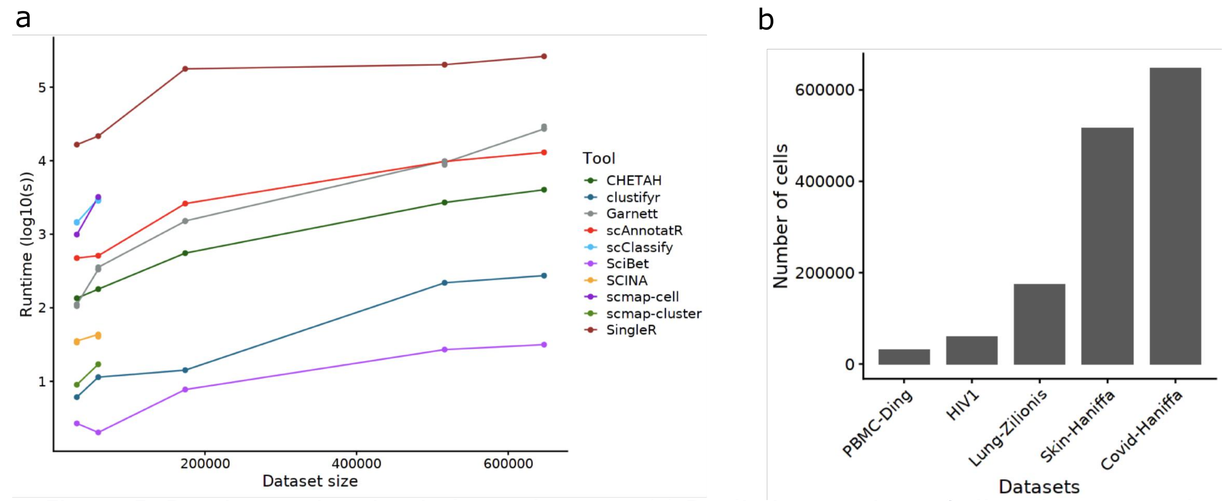

Figure 5: Runtime estimation in large datasets. (a) Prediction runtime of all tools measured over five large datasets. (b) Size of five large datasets in increasing order. SciBet and clustifyr were the fastest evaluated tools, followed by CHETAH, Garnett and scAnnotatR who processed the largest dataset in hours. SingleR was significantly slower than the other tools and needed several days to process the three largest datasets.

Conclusion

scAnnotatR is also among the few tools to provide a dedicated infrastructure to train new cell classifiers. We explicitly provide functions that greatly simplify the training and, most importantly, evaluation of new cell types.